Functional Annotation of *glcE* (PP_3746, UniProt Q88GH7) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

glcE encodes the **FAD-binding subunit of glycolate oxidase (glycolate dehydrogenase), a heterotrimeric, membrane-associated flavoenzyme (GlcD–GlcE–GlcF)** that catalyzes the oxidation of **glycolate (2-hydroxyacetate) to glyoxylate** (EC 1.1.99.14). Rather than reducing O_2, this "oxidase" is mechanistically a **dehydrogenase** that passes electrons from the reduced flavin, via its iron–sulfur subunit (GlcF), into the respiratory electron-transport chain. GlcE itself is the flavin (FAD)-binding module of the complex; it is essential for holoenzyme activity but does not act alone. The enzyme operates in the **cytoplasm/inner-membrane interface** and constitutes the **committed step of glycolate and two-carbon (C2) catabolism**, generating glyoxylate for assimilation via the glyoxylate node (malate synthase / glyoxylate carboligase). In *P. putida* KT2440 this activity is central to metabolizing **ethylene glycol–derived glycolate**, relieving the toxic glycolaldehyde/glycolate bottleneck.

The gene identity is **confirmed and unambiguous**: the *P. putida* KT2440 *glcDEF* operon is directly documented as glycolate oxidase (Franden et al. 2018,

P 29885475

), and the subunit architecture is defined in the classic *E. coli glc* locus studies (Pellicer et al. 1996,

P 8606183

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2. Gene / Protein Identity Verification

Attribute	Value	Source
UniProt accession	Q88GH7	Task metadata
Gene name	*glcE*	UniProt (EMBL AAN69341.1)
Ordered locus	PP_3746	UniProt
Organism	*Pseudomonas putida* KT2440	UniProt
Annotated activity	Glycolate oxidase, putative FAD-binding subunit; EC 1.1.99.14	UniProt
Key domains	FAD-bd_PCMH (IPR016166), FAD-bd_PCMH-like_sf (IPR036318), FAD-bd_PCMH_sub2 (IPR016169), FAD-linked oxidase C (IPR016164), Oxid_FAD_bind_N (IPR006094)	InterPro

Verification outcome — CONFIRMED. All independent lines of evidence agree: - The gene symbol *glcE* corresponds to the FAD/flavin subunit of the **glc** (glycolate) system, exactly matching the UniProt description "Glycolate oxidase, putative FAD-binding subunit." - The organism-specific literature (Franden et al. 2018) explicitly names the *P. putida* KT2440 **glcDEF** operon as glycolate oxidase. - The InterPro **FAD-bd_PCMH** domain set places GlcE in the **PCMH-type FAD-binding / vanillyl-alcohol-oxidase (VAO) flavoprotein superfamily**, the same superfamily as the flavo-subunits of bacterial glycolate/D-lactate oxidoreductases — consistent with an FAD-binding role.

No conflicting "same-symbol, different-gene" literature was found; *glcE* is used consistently across bacteria for this subunit. (Note: eukaryotic/plant "glycolate oxidase," GOX, is an unrelated peroxisomal FMN enzyme of the α-hydroxyacid oxidase family and is **not** the same protein; the bacterial GlcDEF system described here is the correct homolog.)

2.1 Genome-level confirmation (synteny + orthology, this study)

A KEGG genomic-neighborhood query of *P. putida* KT2440 (retrieved 2026-07-16) shows PP_3746 sits in a **canonical, syntenic glycolate-utilization operon** that mirrors the *E. coli glc* locus:

Locus	Gene	KEGG Ortholog (KO)	Annotation
PP_3744	*glcC*	K11474	GlcC glc-operon transcriptional activator (GntR family) — divergently oriented
PP_3745	*glcD*	K00104	glycolate dehydrogenase, **FAD-linked (catalytic) subunit** (EC 1.1.99.14)
PP_3746	***glcE***	**K11472**	glycolate dehydrogenase, **FAD-binding subunit** (EC 1.1.99.14) ← target
PP_3747	*glcF*	K11473	glycolate dehydrogenase, **iron–sulfur subunit**
PP_3748	*glcG*	K11477	glc operon protein GlcG (accessory)

This places GlcE (K11472) alongside its obligate partners GlcD (K00104) and GlcF (K11473) under the divergent GlcC regulator — genome-level corroboration of the biochemical assignment.

2.2 Sequence/domain architecture (this study)

Q88GH7 is a **350-residue** protein bearing a single **FAD-binding PCMH-type domain (residues 1–173; Pfam PF01565 "FAD_binding_4"; InterPro IPR016166/IPR016169/IPR006094; Gene3D 3.30.465.10; SUPFAM SSF56176 + SSF55103)** and **no predicted transmembrane segments**. This is the diagnostic N-terminal FAD-binding fold of the **PCMH/vanillyl-alcohol-oxidase (VAO) flavoprotein superfamily**, which also contains the flavo-subunits of bacterial D-lactate dehydrogenase and related 2-hydroxyacid oxidoreductases. The absence of TM helices indicates GlcE is a **soluble/peripheral flavin-binding subunit** that assembles into the complex rather than an integral membrane protein; membrane association of the complex is provided through the GlcF [Fe-S] subunit and its coupling to the respiratory quinone pool. A crude global alignment gave ~22% identity to *E. coli* GlcE (P52045) and ~26% to GlcD (P0AEP9) — modest but typical for diverged orthologs, with the KO assignment and operon synteny being the decisive evidence.

3. Key Findings with Evidence

3.1 Molecular function: FAD-binding subunit of glycolate oxidase

The founding genetic characterization of the *E. coli glc* locus (Pellicer et al. 1996, P 8606183) established the three-subunit architecture that defines this enzyme family:

> "The proteins encoded from glcD and glcE displayed similarity to several flavoenzymes, the one from glcF was found to be similar to iron-sulfur proteins."

and, demonstrating essentiality:

> "The insertional mutation … in either glcD, glcE, or glcF abolished glycolate oxidase activity, indicating that presumably these proteins are subunits of this enzyme."

Thus **GlcD** is the catalytic flavo-subunit, **GlcE** is the FAD/flavin-binding subunit, and **GlcF** is the iron–sulfur electron-transfer subunit. Loss of any one — including *glcE* — abolishes activity, so GlcE is an **obligate, non-redundant catalytic component**, not an accessory factor. The InterPro FAD-bd_PCMH domains on Q88GH7 corroborate the flavin-binding role at the sequence level.

3.2 Reaction catalyzed and substrate specificity

The holoenzyme catalyzes:

glycolate + **[electron acceptor/respiratory quinone] → glyoxylate** + **[reduced acceptor]** (EC 1.1.99.14)

The physiological substrate is the **two-carbon 2-hydroxymonocarboxylate glycolate**. Substrate identity is reinforced by the cognate uptake system in the same operon: GlcA (the glycolate permease) is specific for **2-hydroxymonocarboxylates — glycolate, L-lactate, and D-lactate** (Núñez et al. 2001, P 11283302):

> "specificity for the 2-hydroxymonocarboxylates glycolate, L-lactate and D-lactate."

The name "oxidase" is historical; the membrane-associated GlcDEF complex functions as a **flavin-dependent, respiratory-chain-linked dehydrogenase** (the flavin is reoxidized by the [Fe-S] subunit feeding electrons to quinone, not directly by O_2), which is why the EC number carries the "99" (acceptor = other/quinone) classification.

3.2.1 Structural inference (AlphaFold, this study)

The AlphaFold DB model **AF-Q88GH7-F1 (v6)** is **very high confidence across all 350 residues** (mean pLDDT **93.8**, median 96.2; 89.4% of residues ≥90). Both structural lobes are confidently modeled — the **N-terminal FAD-binding PCMH domain** (res 1–173, mean pLDDT 93.2) and the **C-terminal FAD-linked-oxidase-like "cap" domain** (res 174–350, mean pLDDT 94.4). This **bilobal architecture is the canonical fold of the PCMH/VAO flavoprotein superfamily**, in which the interdomain cleft accommodates the FAD isoalloxazine and substrate. The uniformly high confidence indicates a **well-folded, ordered flavoprotein subunit** (not disordered or degenerate), and the compact globular fold with no predicted transmembrane helices is consistent with a **soluble/peripheral subunit** that docks with GlcD and the [Fe-S] subunit GlcF (which mediate membrane/respiratory-chain coupling).

3.3 Pathway context: the glyoxylate node and C2 assimilation

Glycolate oxidation is the **committed entry step** into two-carbon metabolism, producing glyoxylate as the pivotal intermediate (Pellicer et al. 1999, P 9880556):

> "The metabolic pathways specified by the glc and ace operons in Escherichia coli yield glyoxylate as a common intermediate, which is acted on by two malate synthase isoenzymes."

Downstream, glyoxylate is assimilated by two routes: - **Malate synthase (GlcB / AceB):** glyoxylate + acetyl-CoA → malate (glyoxylate shunt), and - **Glyoxylate carboligase (Gcl):** 2 glyoxylate → tartronate semialdehyde → glycerate → central metabolism.

In *E. coli* the structural genes are organized as the **glcDEFGB** operon (with *glcG* dispensable) under the divergent **GlcC** activator, dependent on **integration host factor (IHF)** and repressed by the respiratory regulator **ArcA-P** — a regulatory logic that cross-induces the glc and ace systems because both converge on glyoxylate (Pellicer et al. 1999, P 9880556; Pellicer et al. 1996, P 8606183).

3.4 Physiological role in *P. putida* KT2440 — ethylene glycol / glycolate catabolism

In *P. putida* KT2440, ethylene glycol is oxidized through the toxic intermediates **glycolaldehyde → glycolate**, and glycolate oxidase (GlcDEF) is the key enzyme that channels glycolate onward. Franden et al. (2018, P 29885475) showed that engineering efficient ethylene-glycol utilization required relieving the glycolate bottleneck:

> "the additional overexpression of the glycolate oxidase (glcDEF) operon removes the glycolate bottleneck and minimizes the production of these toxic intermediates."

More recent work on *P. putida* strains confirms that robust ethylene-glycol/glycolate metabolism hinges on **glyoxylate assimilation pathways** downstream of glycolate oxidation (Molpeceres-García et al. 2026, P 41453527), reinforcing that GlcDEF-catalyzed glycolate → glyoxylate is the gateway reaction. This makes *glcE* biotechnologically important for **PET/ethylene glycol upcycling** in *Pseudomonas* chassis.

Position in the EG oxidation cascade and downstream regulation. In KT2440, EG is catabolized by a sequence of oxidations **EG → glycolaldehyde → glycolate → glyoxylate**, with **glycolate oxidase (GlcDEF) catalyzing the terminal glycolate → glyoxylate step**. Adaptive laboratory evolution demonstrated that the principal limitation on EG growth is not glycolate oxidation itself but **downstream glyoxylate assimilation and flux balancing**: the key growth-enabling mutation lies in the repressor **GclR**, "which represses the glyoxylate carboligase pathway," and secondary mutations improve growth "by balancing fluxes through the initial oxidations of ethylene glycol to glyoxylate" (Li et al. 2019, P 31166064).

The product glyoxylate is natively assimilated through the **glycerate pathway (glyoxylate carboligase route)**, described as wasteful because it releases CO_2 — which has motivated engineering of alternative routes such as the **β-hydroxyaspartate cycle** to improve EG assimilation (Schada von Borzyskowski et al. 2023, P 36731627).

Thus GlcE, as the FAD-binding subunit of GlcDEF, sits at the **convergence point where all EG/C2 carbon enters central metabolism** via glyoxylate.

3.5 Subcellular localization

The GlcDEF glycolate oxidase is a **membrane-associated (cytoplasmic/inner-membrane) flavoenzyme complex** whose catalytic reaction occurs on the cytoplasmic face; the GlcF iron–sulfur subunit couples flavin reoxidation to the membrane respiratory chain (quinone pool). This localization is inferred from (i) the requirement for a dedicated permease (GlcA) to import glycolate into the cytoplasm before oxidation (P 11283302), and (ii) the general architecture of PCMH/VAO-family FAD + [Fe-S] respiratory dehydrogenases, which are anchored/associated at the inner membrane to donate electrons to quinone. GlcE, as the FAD-binding subunit, functions within this complex at the membrane interface rather than as a soluble standalone enzyme.

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4. Supported and Refuted Hypotheses

Supported: - **H1:** *glcE* encodes the FAD-binding subunit of glycolate oxidase (GlcDEF). — Supported by direct organism-specific naming (P 29885475), subunit genetics (P 8606183), and InterPro FAD-bd_PCMH domains. - **H2:** The enzyme oxidizes glycolate to glyoxylate as its primary reaction. — Supported by EC 1.1.99.14 assignment and pathway genetics (P 9880556 P 8606183). - **H3:** GlcE is obligate for activity (not accessory). — Supported: *glcE* insertional mutants lose all glycolate oxidase activity (P 8606183). - **H4:** The functional role is entry into the glyoxylate node / C2 assimilation, and specifically ethylene-glycol/glycolate catabolism in *P. putida*. — Supported (P 9880556 P 29885475 P 41453527).

Refuted / excluded: - The protein is **not** the eukaryotic peroxisomal FMN glycolate oxidase (GOX, α-hydroxyacid oxidase family). The bacterial GlcDEF system is a distinct PCMH/VAO-family FAD-dependent respiratory dehydrogenase. - GlcE is **not** a soluble monomeric oxidase; it acts only within the GlcD–GlcE–GlcF complex.

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5. Limitations and Future Directions

- **Direct biochemistry is from the *E. coli* ortholog**, not from purified *P. putida* KT2440 GlcE; the *P. putida* evidence is genetic/physiological (operon-level). Subunit-resolved kinetics (kcat/Km for glycolate, flavin type, redox potentials) for the *P. putida* enzyme are not yet reported.
- **Flavin covalency and cofactor stoichiometry** (whether GlcE binds FAD covalently or non-covalently) are inferred from the PCMH/VAO family and not experimentally verified for Q88GH7.
- **Structural data** (experimental or AlphaFold-validated quaternary arrangement of GlcDEF) would clarify how GlcE positions FAD relative to the GlcD active site and the GlcF [Fe-S] chain.

- Future work: heterologous purification and steady-state kinetics of *P. putida* GlcDEF; identification of the physiological electron acceptor (quinone) and membrane topology; site-directed mutagenesis of GlcE flavin-binding residues.

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6. Key References

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